

Functional load modulates acoustic and articulatory variability in English

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Introduction. Variation in speech production has been shown to be modulated by various factors. This study focuses on information-theoretic considerations (Shannon, 1948), which have been found to predict variability in speech (Baker & Bradlow, 2009, *inter alia*). Despite the general finding that perceptual recoverability pressures lead to lower variability on items that are more informative, consideration of articulatory efficiency, which predicts higher variability due to coarticulation, has also been put forth (Xu & Prom-On, 2019). In a parallel piece of literature, the multiple-to-one relation between articulatory synergies and the speech target suggests that certain flexibility could exist between articulation and acoustics (Perrier & Fuchs, 2015), further relaxing the tension of variability between the two domains. This study combines both aspects and examines variability between phonemically-adjacent phoneme pairs in English. Through unsupervised examination of an ultrasound corpus, we show that 1) functionally-loaded phoneme pairs show stronger distinctiveness, and 2) the many-to-one mappings between articulatory synergies and the acoustics (henceforth, *flexibility*) increase such functional effects in acoustics and decrease them in articulation.

Methods. We measured the amount of variability for a phoneme pair (see **Table 1**) as the distance between the two phonemes' respective distributions in the acoustic and articulatory spaces. We used ultrasound/optical lip imaging data and acoustic data from the Tongue and Lips Corpus (Ribeiro *et al.*, 2020). The imaging and acoustic data were mapped to the corresponding articulatory/acoustic spaces through 3D-/1D-CNN autoencoders. The acoustic data was first converted to MFCCs. The autoencoders were then trained to map each video clip/MFCC vector to a 1024D latent space. The decoder component then learned to reconstruct the original data based on the latent vector. After training, the latent spaces then served as the articulatory/acoustic spaces. The flexibility of each phoneme was taken as the amount of uncertainty of the articulation, given the acoustics. This was estimated as the conditional entropy of the articulatory vector given the acoustic vector through a mixture-density network. The functional load of a phoneme pair was calculated as the number of minimal pairs for the phoneme pair in the British National Corpus (BNC Consortium, 2007). Finally, the amount of variability for a phoneme pair was taken as the Maximum Mean Discrepancy between the distributions of all tokens of the two phonemes.

Results. Overall, distinctiveness was modulated by the functional load of the pair. Pairs with higher functional loads tended to have larger distinctiveness. However, such effects also differed between the two modalities, where functional load had positive effects in acoustics but negative effects in articulation. More importantly, flexibility further enhances such a difference in the magnitude of functional load effects on pair distinctiveness between the acoustic and articulatory domains.

Discussion. Our results support previous studies where information-theoretic correlates predict speech variation. The general positive effects of functional load indicate orientation toward perceptual recoverability. Functional load effects, however, were reversed in articulation. Crucially, the flexibility of the phonemes enhanced such a difference in functional load effects. The opposite effects between acoustics and articulation likely suggest the competition between perceptual recoverability and articulatory efficiency. Flexibility between the two domains, in this sense, then serves as a window for both pressures to be simultaneously satisfied.

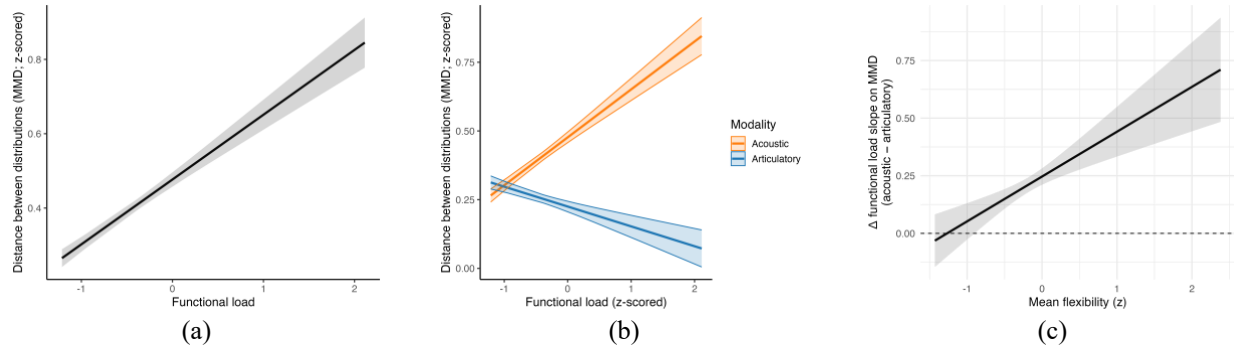


Figure 1: The interaction between the distance between the two phonemes' respective distributions and modality, functional load, and phoneme flexibility.

Table 1: The phonemic pairs examined in this study.

Dimension	Pairs
Vowel height	/i-ɪ/, /e-ɪ/, /ɛ-e/, /ɛ-a/, /a-ɔ/, /u-u/
Place of articulation	/n-ŋ/, /m-n/
Voicing	/k-g/, /b-p/, /t-d/, /ʃ-ʒ/, /s-z/, /f-v/, /tʃ-dʒ/
Liquids and glides	/l-ɹ/, /l-w/, /ɹ-w/

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